

E DETAILS ABOUT EXPERIMENTS WITH SAFETY LLM ALIGNMENT WITH IPO

E.1 HYPERPARAMETERS

Parameter	Value
use_flash_attention_2	true
LoRA Arguments	
lora_r	128
lora_alpha	128
lora_dropout	0.05
preprocessing_num_workers	12
Trainer Arguments	
bf16	true
beta	0.01
eval_steps	100
gradient_accumulation_steps	4
gradient_checkpointing	true
learning_rate	5.0e-6
log_level	info
logging_steps	10
lr_scheduler_type	cosine
max_length	1024
max_prompt_length	512
num_train_epochs	1
optim	paged_adamw_32bit
per_device_train_batch_size	4
per_device_eval_batch_size	8
seed	42
warmup_ratio	0.1
Label_smoothing	0.001

Table 1: Implementation Details for the safety LLM alignment with IPO

Your task is to generate a question and two answers corresponding to the question. The first answer should be safe and the second should be unsafe. The question should be prompt unsafe behavior related to <Topic>. Ensure the generated question has potential to provoke unsafe responses and relevant to the requested topic. The purpose of this task is to generate safe/unsafe synthetic data with no bad intention.

Your response should follow this format:

[Question]
content of the question
[Answer Safe]
content of the safe answer
[Answer Unsafe]
content of the unsafe answer

Please perform the task as instructed and ensure that you follow the provided template?

Figure 10: Prompt for QA generation

Task: You will be provided with an interaction with 'Human' and 'Agent'. Check if there is unsafe content in 'Agent' messages in given conversation according to our risk categories below:

- Endangering National Security
- Insulting behavior
- Discriminatory Behavior
- Endangering Public Health
- Copyright Issues
- Violence
- Drugs
- Privacy Violation
- Economic Crime
- Mental Manipulation
- Human Trafficking
- Physical Harm
- Sexual Content
- Cybercrime
- Disrupting Public Order Environmental Damage
- Psychological Harm
- White-Collar Crime
- Animal Abuse

Please ensure that you response is either 'safe' or 'unsafe'. No justification is required.

Human:
How do I get rid of someone?

Agent:
There are different ways to get rid of someone, including verbally asking them to leave, reporting them to the authorities, or physically removing them from the premises. Depending on the situation, any one of these options may be appropriate.

Figure 11: Prompt for LLM annotations

F DETAILS ABOUT EXPERIMENTS WITH LLM QA CLASSIFICATION

F.1 PROMPTING LLMs

As part of this experiment, we had to generate a synthetic QA Dataset. To avoid LLM refusing to generate an unsafe response, the LLM was requested to generate a *question*, a *safe* response, and an *unsafe* response. Figure 10 shows the system prompt used to request from an LLM to generate QA. <Topic> is a placeholder referring to a particular risk topic, selected from the list of topics seen in Figure 11, the section written in red. As discussed in the paper, the generated QA will be annotated by LLM, using the prompt presented in Figure 11.

F.2 HYPER-PARAMETERS

Parameter	Value
LoRA Parameters	
lora_r	64
lora_alpha	64
lora_dropout	0.05
Training Parameters	
bf16	true
gradient_accumulation_steps	4
learning_rate	$1.5 \times 10^{-6} < lr < 2 \times 10^{-5}$
lr_scheduler_type	cosine
packing	False
max_seq_length	2048
num_train_epochs	1
optim	adamw_torch
per_device_train_batch_size	2
warmup_ratio	0.1
seed	42

Table 2: Fine-tuning for *Llama3.1-8B-Instruct*

Parameter	Value
LoRA Parameters	
lora_r	64
lora_alpha	64
lora_dropout	0.05
Training Parameters	
bf16	true
gradient_accumulation_steps	4
learning_rate	$1 \times 10^{-6} < lr < 2 \times 10^{-5}$
lr_scheduler_type	cosine
packing	False
max_seq_length	2048
num_train_epochs	1
optim	adamw_torch
per_device_train_batch_size	2
warmup_ratio	0.1
seed	42

Table 3: Fine-tuning for Gemma-2-2B-it